

Engaging the Patient in Healthcare:
Conversational Assistants
in the era of LLMs

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Challenges Incarnated

- ❖ Virtual Health Coach: support patients (and human coaches) in adopting SMART goals via SMS exchanges
- ❖ HFChat: conversational agent for Heart Failure patient education

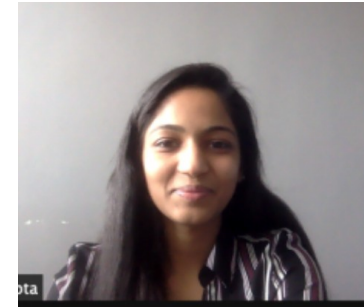
HFChat

DPI Seed award, National Science Foundation 2232307
(2022-2025)

5 Collaborators – 4 colleges



3 Phd Students



HFChat: Conversational agent for African American patients with heart failure (HF)

Hypothesis: African-American (AA) patients who have access to **culturally competent** health care information will make better informed self-care choices

Goals:

1. Uncover self-care concerns of AA patients with HF
2. Explore neuro-symbolic approaches to dialogue agents
3. Evaluate LLMs and take advantage of them if appropriate

Challenge: No formalized method to model and evaluate cultural competence

20 Patient / Educator interactions

Standard Heart Failure (HF) education information comprises:

- ❖ what HF is
- ❖ medications - "water pill" (diuretic)
- ❖ importance of low sodium diet
- ❖ benefits of physical activity
- ❖ daily checks to recognize symptoms
- ❖ value of follow-up appointments

Avg/dialogue	Patient	Educator
Turns	108 (48%)	117 (52%)
Sentences	131 (39%)	205 (61%)
Words	850 (27%)	2280 (73%)

Excerpt of patient-educator conversation [Gupta et al 2020]

---- Patient Control

Patient: Yeah, I don't, I don't do the frozen meal.

Educator: Okay.

Patient: I was basically doing the uh, vegetables.

Educator: Okay.

Patient: Frozen vegetables

---- Educator Control

Educator: They should be fine.

Patient: Yeah.

Educator: But... but, but I do want you to start looking at those nutrition labels.

Patient: Okay.

Educator: So, the other we always want you to do is, um, of course take all your medicines like
you're supposed to.

---- Patient Control

Patient: Which I didn't do last night.

Three prototypes on salt intake

- ❖ HFChat [Gupta et al 2021]: based on [Google DialogFlow](#)
 - Pilot evaluation with 9 patients: Only 32% of questions answered correctly
- ❖ **HFFood-NS** [Tayal et al 24, 26]: NeuroSymbolic architecture
based on PPTOD [Su et al 22]
- ❖ **HFFood-GPT** [Tayal et al 26]: Based on GPT4

HFFood-NS: Synthetic Dataset [Tayal et al 24]

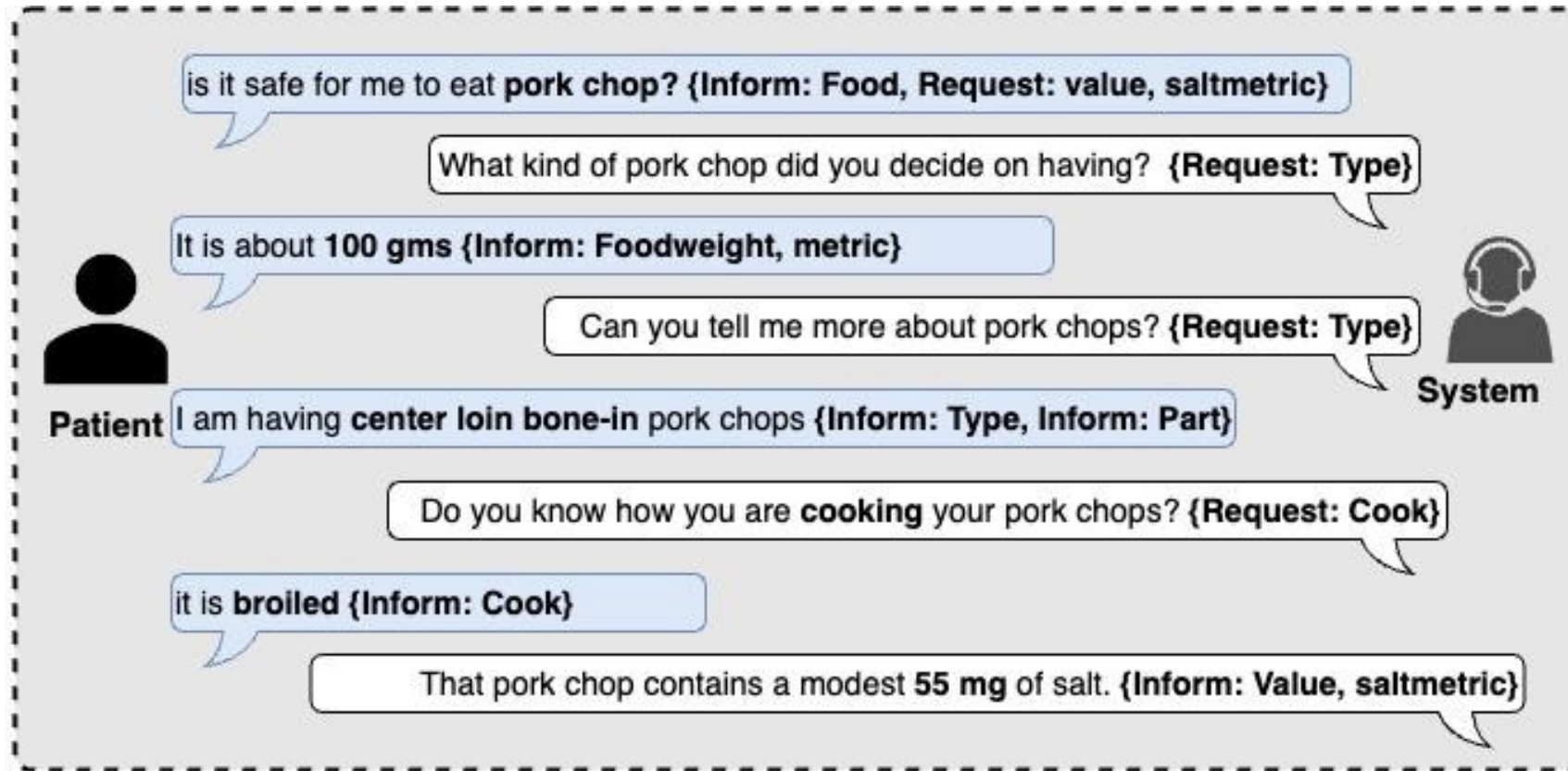
Template based conversational dataset (MultiWOZ [Budzianowski et al 2018])

- ❖ Conversation alternates between user and system
- ❖ **Slots:** include food, cook, type, part, food weight, metric [USDA food DB fields]
- ❖ System questions: one slot at a time; user responses randomly selected from template
- ❖ Types of user responses:

- Matching answers
- Misaligned answers
- Revised answers

# Dialogues	87,425
# Turns	525,392
Avg Turn/Dialogue	6

Synthetic Conversation example



HFFood-NS [Tayal et al SigDIAL 24]

PPTOD (Plug-and-Play Task Oriented Dialog System) [Su et al 22]

- T5 based - designed for task-oriented dialogue
- Trained on synthetic dataset
- Performance: correctly identified most slot values but not salt values

HFFood-NS: Adds neuro-symbolic rules

- Retrieval of accurate salt value from database
- Calculation of correct salt value for queried food weights (based on USDA Food Composition Database)

Translate dialog to belief state: $\{x_t\}$

[usr] Is it safe for me to eat pork chop?
[sys] What kind of pork chop did you decide on having?
[usr] It is about 100 gms.
[sys] Can you tell me more about pork chops
[usr] I am having center loin bone-in pork chops
[sys] Do you know how you are cooking your pork chops?
[usr] it is broiled

PPTOD

[food] {**food**:pork chop, **nutrient**: salt, **foodweight**:100, **metric**: gms, **part**: center loin, **type**: bone-in, **cook**: broiled, **value**: 12, **saltmetric**: mg}

Neuro Symbolic Rules

{value: 55, saltmetric: mg}

Final response:

[sys] That pork chop contains a modest 55 mg of salt.

HFFood-GPT - Prompt

Sodium Scout greets users warmly and helps analyze the salt content in various foods. It methodically asks users about the **food type, cooking method, and portion size, one question at a time**, to accurately determine the salt content.

Using values from a provided JSON file, Sodium Scout calculates the estimate salt content and compares it to the recommended daily intake of 2000mg. It advises that foods exceeding 20% of this intake are not recommended, while those below 5% are favorable choices.

Sodium Scout refrains from giving health advice and suggesting consulting a professional for dietary guidance. Answers are kept **under 40 words**, and it only searches the data provided in the JSON file. **Users do not know about the data file, so don't discuss it. Only focus on information related to food and their salt amount. Do not look for information on the web.**

Within-subject User Study [Tayal et al LREC 26]

- ❖ 20 AA, hospitalized patients interact first with HFFood-NS, then with HFFood-GPT, or viceversa (balanced order)
- ❖ Health Literacy & Digital Literacy were assessed with validated instruments (BRIEF [Haun et al 2009]; eHEALS [Norman & Skinner 2006])
- ❖ Asked 5 questions of first system, same questions of second system
- ❖ Evaluation: system accuracy (salt values, slot values, hallucinations); post-survey questionnaire; readability

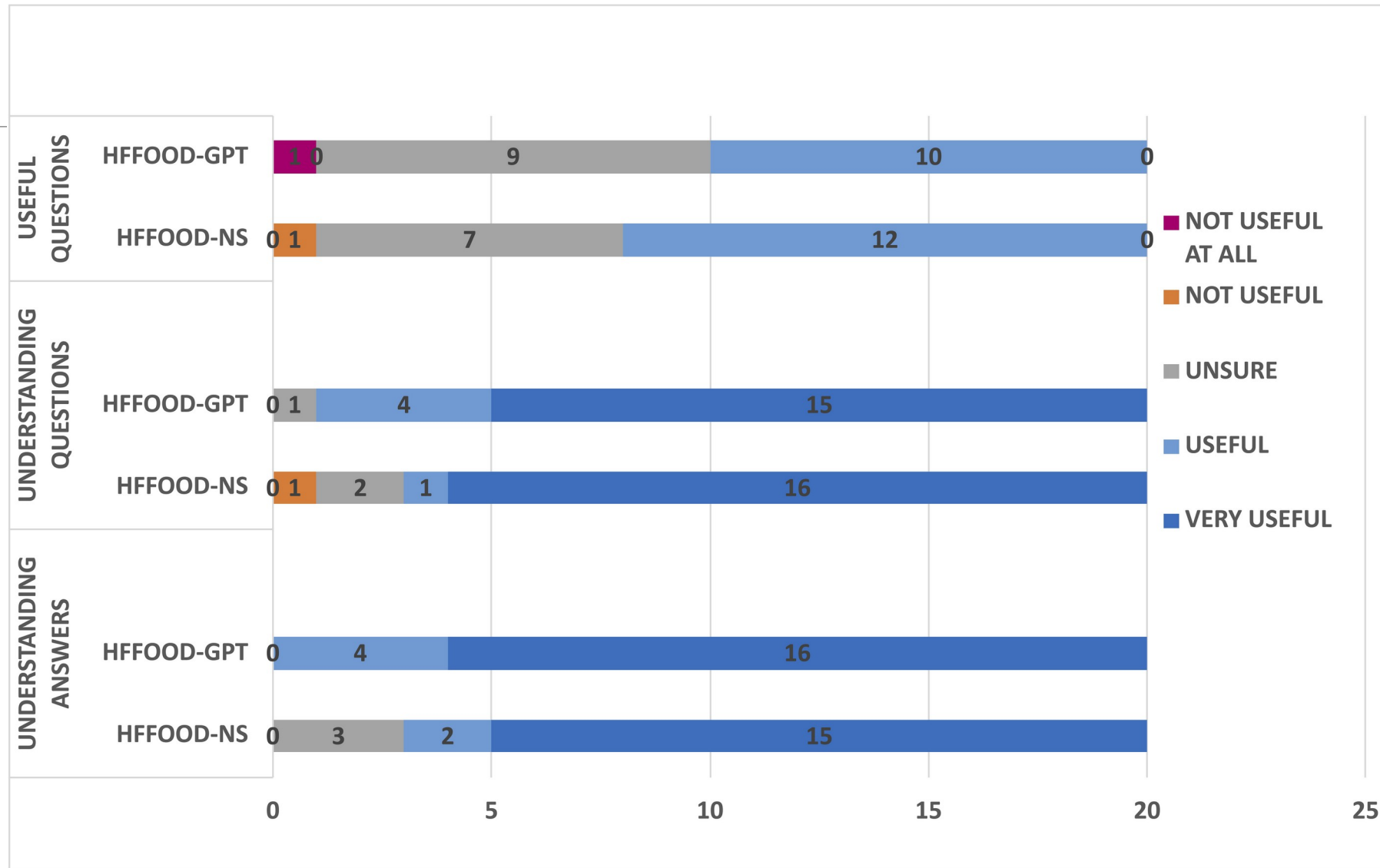
Additional details

- ❖ Systems identified only as Lion (HFChat-NS) / Shark (HFChat-GPT)
- ❖ Balanced presentation: one patient starts with Shark, next with Lion
- ❖ Minimalistic User Interface on tablet:
 - two radio buttons to select system: Shark/Lion
 - one button to indicate whether system is recording or listening
- ❖ Both systems use same Speech Recognizer: OpenAI Whisper-1 [Radford et al 23]
- ❖ HFFood-NS: if slot unfilled asks twice, leaves unfilled after that

HFFood-NS vs HFFood-GPT: intrinsic metrics

	HFFood-NS	HFFood-GPT
Task completion	84%	62%
Salt value accuracy	37%	24%
Slot accuracy	56%	89%

User questionnaire: no preferences



HFFood-NS vs HFFood-GPT: observations

	HFFood-NS	HFFood-GPT
Allows Error Analysis	✓	✗
Reliable	✓	✗
Handles Complex Queries	✗	✓
Fluent	✗	✓
Concise	✓	✗
Faster to deploy	✗	✓

HFFood-GPT Prompt: keep answer max 40 words → avg = 51 words

Complex user queries / unexpected expressions

“What is the sodium content in two grilled hot links with three ounces of barbecue sauce?” (more difficult for HFFood-NS)

Double-dipped Italian beef with cheese and hot giardiniera peppers.

what is the serving size of the Italian beef ?

about the size of a cell phone.

What is your Italian beef portion like ?

The size of a cellphone, fat cellphone.

you will find about nine hundred and fourteen miligrams (46%) of salt in Italian beef. It is around six pinch of salt. You are about to reach half of your daily intake.

Che pedante questo GPT ...

System: Thank you for specifying the preparation method. Now, could you please tell me the portion size of the grilled turkey sausage in grams?

User: 5 milligrams.

System: It seems there might be a misunderstanding with the portion size. Typically, food portion sizes are measured in grams (g) and would be much larger than 5 milligrams (mg). Could you please provide the portion size again in grams? For reference, a common serving size for sausage might be around 50 to 150 grams.

User: 100 milligrams

System: It looks like we're still not quite accurate on the portion size. When we're talking about turkey sausage, the realistic portion size is usually around 100 grams (g), not milligrams (mg). One gram is equal to 1000 milligrams, so a portion size of turkey sausage might be 100 grams (which is 100,000 milligrams), for instance. Could you clarify the portion size in grams for the turkey sausage?

HFFood-NS vs HFFood-GPT: Response readability

Healthcare information should be at 6th grade reading level (prima media)

US grades: 1-12

US school grade level:

SMOG better for healthcare
[Wong et al 13, Mac et al 22]

- SMOG [McLaughlin 69]:

$$3 + \sqrt{\text{polysyllabic word count}}$$

- FKGL [Kincaid et al 75]:

$$0.4 * \frac{\text{Total Words}}{\text{Total Sentences}} + 12 * \frac{\text{Total Syllables}}{\text{Total Words}} - 16$$

HFFood-NS vs HFFood-GPT: Response readability

Reading ease:

- FKRE [0..100] [Flesh 48, Kincaid et al 75]

$$206.835 - 1.015 * \frac{\text{Total Words}}{\text{Total Sentences}} - 84.5 * \frac{\text{Total Syllables}}{\text{Total Words}}$$

	HFFood-NS	HFFood-GPT
SMOG (↓) [grade]	7.05	10.58
FKGL (↓) [grade]	2.71	7.10
FKRE (↑) [ease]	91.21	71.79

HFCHat: take aways

- ❖ **Dataset:** collected 20 patient/educator conversations (not sharable); generated synthetic dataset
- ❖ **Bias / fairness:** meant to address cultural competence for AA (Hispanics too). To be continued
- ❖ **Neurosymbolic** approaches: developed **HFFood-NS**
- ❖ **LLMs:** currently, uncertain utility
 - a. Doesn't follow prompt instructions; hallucinations
 - b. Too high reading level

Current and Future Work (especially HFChat)

- ❖ Cultural Competence: "you know when you see it" →
Patient-centered care: empathy, respect, curiosity, responsiveness
- ❖ Health-belief model [Hochbaum 1958, Rosestock 2000] & Persuasiveness:
perceived [susceptibility to risk; risk severity; benefits of action,
barriers to action]; self-efficacy; cues to action
- ❖ Extend to other domains – via schema graphs used to control LLMs
[Agrawal et al SigDIAL 24, Zhang et al EMNLP 23, Zhao et al EMNLP 23]
- ❖ Bias ...

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